

Replication Package

"Peer Effects and Rank Concerns in the Classroom"

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Replication Package Version: March 27 2026

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1. Overview

This replication package reproduces all tables, figures, and in-text numerical results reported in:

Tincani, Michela (2026). "Peer Effects and Rank Concerns in the Classroom", conditionally accepted at the *Journal of Political Economy*.

The package contains:

- All code necessary to reproduce the empirical results.

- All raw datasets that the author is permitted to redistribute.
- Instructions to regenerate all tables, figures, and in-text numbers.

The folder **output/** should be empty when received by replicators.
All results are generated by running the code as described below.

2. Structure of the Replication Package

```
Replication_Package_Tincani-20180648
├── code
│   ├── excel
│   ├── python
│   ├── R
│   ├── stata
│   └── ado_frozen
├── tex
├── data
│   ├── raw
│   └── processed
├── confidential-data-not-for-publication
│   ├── raw
│   └── processed
├── output
│   ├── figures
│   ├── tables
│   └── in_text_numbers
├── README.md
├── README.pdf
└── Manuscript_Replication_Package_Tincani-20180648.pdf
```

Notes

- All files inside **output/**, **data/processed**, and **confidential-data-not-for-publication/processed** are generated by the replication scripts.
- The folder **data/raw** contains all data inputs that the author is permitted to redistribute publicly. These include author-constructed datasets and any third-party data for which redistribution is allowed.
- The folder **confidential-data-not-for-publication/raw** is intended to store restricted administrative datasets that cannot be publicly redistributed. In the publicly deposited replication package, this folder will be empty. Researchers seeking to replicate the results must independently obtain the relevant restricted datasets from the originating institutions and place them in this folder.
- The folder **code/stata** contains all Stata do-files used in the analysis, the subfolder **ado_frozen/** containing the frozen user-written Stata ado dependencies used by the scripts, and the file **eb_map_iv_f.txt**, which contains starting values for the latent class (LCA) estimation. The starting values are provided to ensure numerical reproducibility of the estimation procedure.
- The README is provided in both Markdown (README.md) and PDF (README.pdf) formats; the contents are identical.

3. Data Availability and Provenance Statement (DAS)

3.1 Statement About Rights

The author has legitimate access to and permission to use all data employed in this manuscript.

Some datasets originate from Chilean public institutions:

- Instituto Nacional de Estadísticas (INE), Chile
- Ministerio de Educación, Chile
- Agencia de Calidad de la Educación, Chile

The author does **not** hold redistribution rights for these datasets beyond the permissions granted by the respective institutions. Full replication requires independently obtaining the restricted datasets listed in Section 3.2.

3.2 Data Inventory

Files located under **data/raw/** are included in the replication package. All other datasets must be obtained from their original source.

File(s)	Source	Description	Access
---------	--------	-------------	--------

File(s)	Source	Description	Access
20130425_Rendimiento_2005_20060316_PUBL.csv, 20130226_Rendimiento_2007_20080312_PUBL.csv, Rendimiento por estudiante 2008.csv, 20130227_Rendimiento_2009_20100715_PUBL.csv, Rendimiento por estudiante 2010.csv, 20130301_Rendimiento_2011_20120416_PUBL.csv	Ministerio de Educación, Chile	Rendimiento por Estudiante administrative dataset: student- and school-level records on grades, student gender, school enrollment, years 2005, 2007, 2008, 2009, 2010, 2011.	Publicly available; see note (1).
data/raw/20140805_matricula_unica_2009_20090430_PUBL.csv	Ministerio de Educación, Chile	Matrícula por Estudiante administrative dataset: student- and school-level records on enrollment, year 2009.	Publicly available; see note (2).
data/raw/CPV2002_PERSONA.csv, data/raw/CPV2002_Vivienda.csv	Instituto Nacional de Estadísticas (INE), Chile	2002 Chile Population and Housing Census (persons and dwellings modules)	Public microdata; official access via INE application process (https://www.ine.gob.cl/); downloaded in 2020 and 2021.
BBDD Alumnos 4to 2005.csv, BBDD Alumnos 4to 2007.csv, 2009 8b bbdd ptje estudiante.csv, ALUM8VO_2011 mrun.csv	Agencia de Calidad de la Educación, Chile	Puntajes por estudiante administrative dataset: standardized test score data for fourth grades in 2005 and 2007, and for eighth grades in 2009 and 2011.	Restricted; access subject to Agency approval (see https://informacionestadistica.agenciaeducacion.cl/#/bases).
SIMCE_8B_2009_CEST.csv, CEST8VO.csv	Agencia de Calidad de la Educación, Chile	Respuestas a Cuestionarios Estudiantes administrative dataset: answers to student questionnaire, with information on effort cost and course engagement, for eighth graders in 2009 and 2011.	Restricted; access subject to Agency approval (see https://informacionestadistica.agenciaeducacion.cl/#/bases)

File(s)	Source	Description	Access
2007 4b bbdd cpadres.xlsx, SIMCE_4b_2005_CPADRES.xlsx, 2009 8b bbdd cpadres.csv, CPA8V0.xlsx	Agencia de Calidad de la Educación, Chile	Respuestas a Cuestionarios Apoderados administrative dataset: answers to parental questionnaire, with information on household characteristics, for parents of fourth graders in 2007 and 2005, and for parents of eighth graders in 2009 and 2011.	Restricted; access subject to Agency approval (see https://informacionestadistica.agenciaeducacion.cl/#/bases).
2009 8b bbdd cprofesor leng.xlsx, 2009 8b bbdd cprofesor mate.xlsx, CPRL8V0.xlsx, CPRM8V0.xlsx	Agencia de Calidad de la Educación, Chile	Respuestas a Cuestionarios Docentes: answers to questionnaire for Spanish and Mathematics teachers, with information on curriculum covered, for teachers of eighth graders in 2009 and 2011.	Restricted; access subject to Agency approval (see https://informacionestadistica.agenciaeducacion.cl/#/bases).
data/raw/GSHH_f_L1.shp, data/raw/GSHH_f_L1.shx	National Centers for Environmental Information (NOAA)	Global Self-consistent Hierarchical High-resolution Geography (GSHHG) shoreline shapefiles (for Chile coastline)	Publicly available; see note (3).
data/raw/earthquake_catalog_6.csv	United States Geological Survey (USGS)	Global earthquake events (Magnitude ≥ 6), worldwide, January 1900–September 2023	Publicly available; see note (4).
data/raw/EE_cerr_201011.csv	Author (derived from Ministerio de Educación records)	Schools closed due to the 2010 earthquake and schools receiving displaced students (2010-2011)	Author-constructed dataset derived from restricted ministerial administrative records. Included in replication package.

File(s)	Source	Description	Access
data/raw/comuna_alu_geocoded.csv	Author	Municipality geographic coordinates with administrative identifiers	Included in replication package.
data/raw/sch_com_dist.csv	Author	School-town-to-asperity distance measures	Included in replication package.
data/raw/stud_com_dist.csv	Author	Student-town-to-asperity distance measures	Included in replication package.
data/raw/expcat_consulting.csv , data/raw/expcat_other.csv	Author	Aggregated expenditure category groupings	Included in replication package.
SS_Rendidos_2008-2010_20120704.dta	Ministerio de Educación, Chile	SEP school expenditure reports	Restricted administrative dataset provided by Ministerio de Educación; access subject to Ministry approval; see note (5).
data/raw/ne_110m_admin_0_countries.zip	Natural Earth	Natural Earth 1:110m scale Cultural Vectors - Admin 0 Countries; used to draw country boundaries (Chile map base layer).	Publicly available; see note (6).

Notes

- (1) Rendimiento por Estudiante dataset available at <https://datosabiertos.mineduc.cl/rendimiento-por-estudiante-2/>. The analysis uses the version accessed in 2013 (for the years 2005, 2007, 2009, 2011), in 2018 (for the year 2010), and in 2021 (for the year 2008). Subsequent updates or revisions to the datasets available through the portal may differ from the versions used in this study.
- (2) Matrícula por Estudiante dataset avialable from the Ministry of Education open data portal (<https://datosabiertos.mineduc.cl/matricula-por-estudiante-2/>). The analysis uses the version accessed in 2014. Subsequent updates or revisions to the datasets available through the portal may differ from the versions used in this study.
- (3) Downloaded from <https://www.ngdc.noaa.gov/mgg/shorelines/data/gshhg/latest/> on 28 March 2025.
- (4) USGS Earthquake Catalog; query: magnitude ≥ 6 , worldwide, 1900–September 2023; downloaded from <https://earthquake.usgs.gov/earthquakes/search/> on 9 September 2023.
- (5) The version shared confidentially with the JPE Data Editor is redacted of one variable containing written descriptions of single expenditure items, as per approved exemption.
- (6) Downloaded from https://naciscdn.org/naturalearth/110m/cultural/ne_110m_admin_0_countries.zip on 24 February 2026.

3.3 Data Citations

Agencia de Calidad de la Educación, Chile. n.d. *Base de Datos Puntajes SIMCE por Estudiante (4° básico 2005, 2007; 8° básico 2009, 2011)* [dataset]. Agencia de Calidad de la Educación. Restricted administrative data; access subject to Agency approval. <https://informacionestadistica.agenciaeducacion.cl/#/bases>

Agencia de Calidad de la Educación, Chile. n.d. *Cuestionarios Apoderados SIMCE (4° básico 2005, 2007; 8° básico 2009, 2011)* [dataset]. Agencia de Calidad de la Educación. Restricted administrative data; access subject to Agency approval. <https://informacionestadistica.agenciaeducacion.cl/#/bases>

Agencia de Calidad de la Educación, Chile. n.d. *Cuestionarios Docentes SIMCE Lenguaje y Matemática (8° básico 2009, 2011)* [dataset]. Agencia de Calidad de la Educación. Restricted administrative data; access subject to Agency approval. <https://informacionestadistica.agenciaeducacion.cl/#/bases>

Agencia de Calidad de la Educación, Chile. n.d. *Cuestionarios Estudiantes SIMCE (8° básico 2009, 2011)* [dataset]. Agencia de Calidad de la Educación. Restricted administrative data; access subject to Agency approval. <https://informacionestadistica.agenciaeducacion.cl/#/bases>

Instituto Nacional de Estadísticas (INE), Chile. 2002. *Censo de Población y Vivienda 2002: Microdatos (Personas y Viviendas)* [dataset]. Instituto Nacional de Estadísticas. Access via official INE application process; accessed 2020 and 2021. <https://www.ine.gob.cl/>

Ministerio de Educación, Chile. n.d. *Matrícula por Estudiante, 2009* [dataset]. Datos Abiertos MINEDUC. Downloaded 2014. <https://datosabiertos.mineduc.cl/matricula-por-estudiante-2/>

Ministerio de Educación, Chile. n.d. *Rendimiento por Estudiante, 2005–2011* [dataset]. Datos Abiertos MINEDUC. Years 2005, 2007, 2009, 2011 downloaded 2013; year 2010 downloaded 2018; year 2008 downloaded 2021. <https://datosabiertos.mineduc.cl/rendimiento-por-estudiante-2/>

Ministerio de Educación, Chile. n.d. *SEP School Expenditure Reports (SS_Rendidos_2008–2010_20120704)* [dataset]. Ministerio de Educación. Restricted administrative data; accessed 2012; access subject to Ministry approval.

National Centers for Environmental Information (NOAA). n.d. *Global Self-consistent Hierarchical High-resolution Geography (GSHHG), latest version* [dataset]. NOAA. Downloaded 28 March 2025. <https://www.ngdc.noaa.gov/mgg/shorelines/data/gshhg/latest/>

Natural Earth. n.d. *Natural Earth 1:110m Cultural Vectors – Admin 0 Countries* [dataset]. Natural Earth. Downloaded 24 February 2026. https://naciscdn.org/naturalearth/110m/cultural/ne_110m_admin_0_countries.zip

Tincani, Michela M. 2026. *Aggregated School Expenditure Category Groupings* [dataset]. Author-constructed research dataset; included in replication package.

Tincani, Michela M. 2026. *Geocoded Municipality Coordinates and Distance Measures* [dataset]. Author-constructed research dataset; included in replication package.

Tincani, Michela M. 2026. *School-Town-to-Asperity and Student-Town-to-Asperity Distance Measures* [dataset]. Author-constructed research dataset; included in replication package.

Tincani, Michela M. 2026. *Schools Closed Due to the 2010 Chile Earthquake and Receiving Schools (2010–2011)* [dataset]. Author-constructed dataset derived from restricted Ministerio de Educación administrative records; included in replication package.

United States Geological Survey. n.d. *USGS Earthquake Catalog (Magnitude ≥ 6 , Worldwide, 1900–September 2023)* [dataset]. USGS. Downloaded 9 September 2023. <https://earthquake.usgs.gov/earthquakes/search/>

4. Software Requirements

4.1 Operating System

Tested on:

- Windows 10

4.2 Python

- Python version 3.12 (tested: 3.12.10)

Required Packages

For `calculate_coastal_proximity.py`:

```
python -m pip install pandas
python -m pip install geopandas
python -m pip install geopy
```

For `create_map.py`:

```
python -m pip install matplotlib
python -m pip install geodatasets
```

Using `python -m pip` ensures that packages are installed in the same Python environment used to run the scripts.

Versions of packages

```
pandas==2.3.3
numpy==2.3.4
geopandas==1.1.1
pyogrio==0.11.1
pyproj==3.7.2
shapely==2.1.2
geopy==2.4.1
matplotlib==3.10.7
geodatasets==2024.8.0
```

Additional dependencies (installed automatically):

```
pytz==2025.2
tzdata==2025.2
```

```
six==1.17.0
packaging==25.0
certifi==2025.10.5
geographiclib==2.1
contourpy==1.3.3
cycler==0.12.1
fonttools==4.60.1
kiwisolver==1.4.9
pillow==12.0.0
pyparsing==3.2.5
pooch==1.8.2
platformdirs==4.5.0
requests==2.32.5
charset_normalizer==3.4.4
idna==3.11
urllib3==2.5.0
```

4.3 Stata

Versions Used

- StataNow 19.5 (all do-files except one)
- Stata 15 (only for 21_conley.do)

All do-files specify version 18, except:

- 21_conley.do, which specifies version 15
- 9_lca_all.do, which specifies version 15 to execute gsem

Required Packages

The replication package includes a frozen local copy of the required user-written ado files in:

```
code/stata/ado_frozen
```

The master do-file code/stata/0_Peer_Effects_and_Rank_Concerns.do adds this directory to Stata's ado-path and runs against these local copies.

The frozen Stata dependencies are:

Functionality used in the code	Frozen file(s)	Source package / note	Frozen version/date
esttab	esttab.ado	estout	2.0.6, 02jun2014
estout	estout.ado	estout	3.17, 02jun2014
estadd	estadd.ado	estout	2.3.3, 28may2014
estpost	estpost.ado	estout	1.1.6, 06aug2010
eststo	eststo.ado, _eststo.ado	estout	1.1.0, 05nov2008; 1.0.4, 09nov2007
egen ... = corr(...), covariance	_gcorr.ado	egenmore	09jun2005
egen ... = rmiss2(...)	_grmiss2.ado	dm31 / rmiss2	2.0.0, 24jul1991
ols_spatial_HAC	ols_spatial_HAC.ado	bundled with replication package	header: 6/2010; V3 update 6/2018

4.4 R

- R version 4.5.2 (2025-10-31 ucrt)

Required packages and associated libraries

```
ggplot2, dplyr
```

Automatically installed by:

```
code/R/plot_histogram_earthquakes.R
```

Versions of packages

```
ggplot2: 4.0.2  
dplyr: 1.2.0
```

5. Instructions for Replicators: Order of Execution

Step 0 — Ensure Output Folder Is Empty

Confirm that:

- `output/figures`
- `output/tables`
- `output/in_text_numbers`

are empty before running any code. If `output` is not empty, any existing content may be overwritten when the code is executed.

Step 1 — Python Preprocessing

Navigate to:

```
code/python
```

Install the required packages as per Section 4.2. Then run:

```
python calculate_coastal_proximity.py
```

The script automatically detects the replication package root directory. It generates the intermediate dataset containing the distance of each town from the Chilean coastline. This dataset is used by subsequent scripts.

Step 2 - Stata Data Construction and Analysis

Open Stata and set the working directory to the root of the replication package (i.e., the folder that contains `code/`, `data/`, `output/`, and `confidential-data-not-for-publication/`).

Example:

```
cd "path/to/Replication_Package_Tincani-20180648"
```

Then run:

```
do "code/stata/0_Peer_Effects_and_Rank_Concerns.do"
```

Note: The master do-file `0_Peer_Effects_and_Rank_Concerns.do` defines all internal paths relative to the current working directory (using `c(pwd)`). For this reason, replication requires that Stata's working directory is set to the replication package root before executing the master do-file.

This do-file:

- Loads the frozen Stata ado files stored in `code/stata/ado_frozen`
- Runs all Stata programs
- Generates intermediate and final datasets
- Generates tables and figures
- Produces log files containing in-text numbers

Step 3 — Python Map Creation

Navigate to:

```
code/python
```

Install the required Python packages as described in Section 4.2.

Then run:

```
python create_map.py
```

The script automatically detects the replication package root directory. It generates Figure 7, the map showing the spatial distribution of damages among students' families by socioeconomic status.

Step 4 — R Code

Navigate to:

```
code/R
```

Then run:

```
Rscript plot_histogram_earthquakes.R
```

The script automatically detects the replication package root directory. It installs the required packages, and generates Figure 2, the histogram of worldwide earthquakes with magnitude greater than or equal to 6.

This step can be run independently of all other steps.

6. Instructions for Replicators: Manual Verification Steps

6.1 Figure 3 — Donut Graph (Excel-Based)

The donut graph is constructed in Excel using values generated by Stata.

The numerical inputs are taken from:

```
output/figures/lca_all.log
```

The Excel file used to construct the donut graphs is:

```
code/excel/donut_graph_construction_types.xls
```

The Excel file is already populated. Replicators must verify that the values match those in the log file, as per the instructions reported in Sections from 6.1.1 to 6.1.4. For convenience, the donut.png file that contains the Figure to be included in the paper has been saved in the `code\excel` folder. Once the manual verification step is complete, replicators must move this file to the `output\figure` folder to compile the paper.

6.1.1 Class Fractions (Graph Titles)

The titles of the three donut charts report class fractions. These must match the mean values reported in the log file immediately after:

```
sum cpr*
```

Excel Cell	Corresponding Log File Value
B1	Mean of <code>cpr1</code>
B32	Mean of <code>cpr2</code>
B63	Mean of <code>cpr3</code>

6.1.2 Column C: Posterior-Weighted Proportions

Column C contains posterior-weighted proportions by material type and class.

Rules:

- Values must be copied directly from the log file.
- If a material number (listed in Column D) does not appear in the log file, set the value to `0`.
- No transformations or recalculations should be performed.

Values correspond to:

```
v4a_use | class k
v4b_use | class k
v4c_use | class k
```

for classes ($k = 1, 2, 3$).

6.1.3 Cell–Log Correspondence

Excel Cells	Log File Section	Class
C3–C9	v4a_use	1
C11–C19	v4b_use	1
C21–C29	v4c_use	1
C34–C40	v4a_use	2
C42–C50	v4b_use	2
C52–C60	v4c_use	2
C65–C71	v4a_use	3
C73–C81	v4b_use	3
C83–C91	v4c_use	3

6.1.4 Final Verification Checklist

- ☐ Cells B1, B32, B63 match `sum cpr*`
- ☐ Column C matches log posterior-weighted proportions
- ☐ Missing material numbers coded as 0
- ☐ No manual adjustments performed

Once verified, the donut graphs are fully reproducible.

6.2 Conley Standard Errors (Table A14)

`code/stata/21_conley.do` produces two outputs:

- 1. A LaTeX file:

```
output/tables/conley_to_edit.tex
```

This file serves as a **compilation placeholder** that allows the main LaTeX document to compile successfully.

The table generated from `conley_to_edit.tex` is **not** the final version of Table A14 as shown in the published PDF.

- 2. A log file:

```
output/tables/Table_A14_conley.txt
```

This log file contains the full Conley regression output, including the Conley standard errors.

To verify reproducibility of Table A14, replicators must:

- Open `Table_A14_conley.txt`
- Locate the Conley standard errors reported in the regression output, after the `ols_spatial_HAC` command. These are the standard errors of the variables `post_st_exp`, `post_loo_mean_damage`, `pos_loo_sd_damage`, which are the first three in the regression tables. Variable `ts` is the test score output, variable `GPA_std` is the GPA output.
- Confirm that these standard errors and the p-values match the standard errors and significance stars reported in the published PDF version of Table A14

The authoritative source for verification of the Conley standard errors is the log file `Table_A14_conley.txt`, not the placeholder LaTeX file.

7. Instructions for Replicators: Compiling the Paper

After all outputs are generated:

Move the entire `output/` folder into:

```
code/tex/
```

The LaTeX source uses relative paths that expect this structure.

Alternatively, modify LaTeX paths manually.

8. Computational Requirements and Execution Environment

This section documents the computing environments used to prepare the replication package.

The replication consists of two computational environments:

1. Local execution (all Stata do-files except Conley, all Python files, the R file)
2. Cluster execution (Stata file 21_conley.do)

8.1 Local Execution Environment (Main Pipeline)

All do-files except 21_conley.do were executed locally using:

- **StataNow 19.5**
- Do-files specify **version 18**

Hardware

The following hardware was used for executing the two Python file, the R file, and all do-files except 21_conley.do, :

- Processor: Intel® Core™ Ultra 7 268V 2.20 GHz
- RAM: 32.0 GB (31.6 GB usable)
- System type: 64-bit operating system, x64-based processor

Runtime

- Full Stata pipeline (excluding Conley): around 7 hours

This includes:

- Data preparation
 - Estimation
 - Generation of all tables (except A1, A19, A20)
 - Generation of all figures (except Figures 1, 2, 7, 12, A1)
 - Generation of all in-text number logs
- R file: less than a minute
 - Python file `calculate_coastal_proximity.py`: less than 10 minutes
 - Python file `create_map.py`: less than a minute

No high-performance computing (HPC) resources were required for this portion.

8.2 Cluster Execution (Conley Standard Errors)

The file:

```
code/stata/21_conley.do
```

was executed on the **UCL Myriad HPC cluster**, using **Stata 15**.

The do-file specifies:

```
version 15
```

Cluster Environment

The following modules were loaded:

- beta-modules
- gcc-lib/9.2.0
- libpng/1.6.37/gnu-9.2.0
- stata/15

Other loaded system modules:

- gerun
- emacs/28.1
- userscripts/1.5.0
- rcps-core/1.0.0
- mpi/intel/2018/update3/intel

Stata was run in batch mode.

Job Submission Parameters

The Conley job was submitted using Sun Grid Engine with:

- Memory requested: 188 GB
- Temporary filesystem: 300 GB
- OpenMP threads requested: 1

Execution command:

```
stata -b do conley_10.do
```

The working directory was located in scratch space, as required by Myriad compute nodes.

On the UCL Myriad cluster, the Conley standard errors were executed as **separate batch jobs**, each corresponding to a single distance bandwidth (in km):

- 10
- 25
- 50
- 250 (test score outcome only)
- 350 (GPA outcome only)

The computing environment and resource requests documented above (modules, Stata 15, and the SGE resource requests) are therefore sufficient to reproduce **each bandwidth-specific Conley run separately**.

The do-file `21_conley.do` runs these bandwidth-specific Conley estimations **sequentially** within one script. Running all bandwidths sequentially in a single job may require higher total memory than running them as separate jobs, depending on how intermediate objects are handled during execution.

Runtime

- Approximately 61 hours

Note on set processors and version 18: The do-file `0_Peer_Effects_and_Rank_Concers.do` includes a `set processors 1` that is only supported in Stata/MP and a `version 18` command that is only supported in STATA 18 or higher. On UCL Myriad, Conley estimates were executed using Stata/SE, where `set processors` is not allowed, and with Stata version 15, where `version 18` is not allowed. The replication code therefore guards these command so the do-file runs under both Stata/SE version 15 and Stata/MP version 19.5.

8.3 Memory Requirements

- Main pipeline: runs with 32 GB RAM on local machine.
- Conley estimation: requires high memory allocation (requested 188 GB on cluster).

Users attempting to run `21_conley.do` locally may experience memory limitations depending on dataset size and spatial bandwidth parameters.

8.4 Reproducibility Note

The Conley estimation (`21_conley.do`) was run in Stata 15.

All other do-files were run in StataNow 19.5.

Because `version` is explicitly specified in all do-files, results should be reproducible across compatible Stata versions, subject to:

- Identical random seeds
- Identical sort seeds
- Identical module/library versions
- USe of the bundled frozen ado files

9. License for Code

The original code written by the author and included in this replication package is licensed under:

[MIT License](#)

This license applies only to the original code files authored by the author.

Third-party datasets included in the replication package remain subject to the terms and conditions of their respective data providers.

10. Software Citations

- StataNow 19.5
- Stata 15 (for Conley estimation)
- Python 3.12.10
- R 4.5.2 (2025-10-31 ucrt)

11. Mapping of Exhibits to Code and Output Files

The tables below describe how each exhibit in the paper is generated and which program produces it.

Unless otherwise noted, all exhibits are generated automatically by running:

```
code/stata/0_Peer_Effects_and_Rank_Concerns.do
```

The R and Python programs are explicitly indicated where relevant.

All figures are saved in `output/figures/`.

All tables are saved in `output/tables/`.

11.1 Figures

Output in Paper	Output in Package	Script/Program to Execute	Comments
Figure 1	<code>code/tex/Paper/data.tex</code>	—	Conceptual drawing (tikz figure); not generated from data
Figure 2	<code>output/figures/earthquake_distribution.png</code>	<code>code/R/plot_histogram_earthquakes.R</code>	
Figure 3	<code>output/figures/lca_all.log</code> + Excel file	<code>code/stata/9_lca_all.do</code> + <code>code/excel/donut_graph_construction_types.xls</code>	Manual Excel verification required (see Section 6.1)
Figure 4	<code>output/figures/class_prob_by_p_educ_all.png</code>	<code>code/stata/14_analyze_census.do</code>	
Figure 5	<code>output/figures/fit_vc_prediction_all.png</code>	<code>code/stata/14_analyze_census.do</code>	
Figure 6	<code>output/figures/damages_peduc.png</code>	<code>code/stata/15_descriptive_tables_figures.do</code>	
Figure 7	<code>output/figures/damage_propagation_by_SES_py.png</code>	<code>code/python/create_map.py</code>	
Figure 8	<code>output/figures/heteffects_ts_GPA_by_simce_nofe.png</code>	<code>code/stata/17_het_effects.do</code>	
Figure 9	<code>output/figures/heteffects_ts_GPA_by_simce_fe.png</code>	<code>code/stata/17_het_effects.do</code>	
Figure 10	<code>output/figures/consulting_other_2010_details.png</code>	<code>code/stata/19_school_resources.do</code>	
Figure 11	<code>output/figures/heteffects_GPAranksd_bysimce.png</code>	<code>code/stata/18_med_relationships.do</code>	
Figure 12	<code>code/tex/Paper/classroomA_additive_damage.png</code> , <code>code/tex/Paper/classroomB_additive_damage.png</code> , <code>classrooms_AB_post_earthquake_additive_damage.png</code>	—	Conceptual drawing; not generated from data
Figure A1	<code>code/tex/Paper/reconstruccion_dios.png</code>	—	Field photograph from Comerio (2013); not generated from data

Output in Paper	Output in Package	Script/Program to Execute	Comments
Figure A2	output/figures/heteffects_ts_GPA_by_GPA4_nofe.png	code/stata/17_het_effects.do	
Figure A3	output/figures/heteffects_GPArankmean_bysimce.png	code/stata/18_med_relationships.do	
Figure A4	output/figures/eff_cost_simce.png	code/stata/15_descriptive_tables_figures.do	

11.2 Tables

Output in Paper	Output in Package	Script/Program to Execute	Comments
Table 1	output/tables/descriptive_pre_post.tex	code/stata/15_descriptive_tables_figures.do	
Table 2	output/tables/main_effects_ts_GPA.tex	code/stata/16_main_effects_and_rob.do	
Table 3	output/tables/effects_on_switches.tex	code/stata/20_identifying.do	
Table 4	output/tables/main_effects_ts_2025_lagged.tex	code/stata/20_identifying.do	
Table 5	output/tables/test_identif_assmpt.tex	code/stata/20_identifying.do	
Table 6	output/tables/het_effects_ts_GPA_bysimce.tex	code/stata/17_het_effects.do	
Table 7	output/tables/med_effects_curriculum_2025.tex	code/stata/18_med_relationships.do	
Table 8	output/tables/med_effects_exp114.tex	code/stata/19_school_resources.do	
Table 9	output/tables/med_effects.tex	code/stata/18_med_relationships.do	
Table A1	code/tex/Paper/appendix.tex	—	Conceptual Table; not generated from data
Table A2	output/tables/distrib_materials.tex	code/stata/14_analyze_census.do	
Table A3	output/tables/descriptives_treatvars_damage.tex	code/stata/15_descriptive_tables_figures.do	
Table A4	output/tables/correlates_pre_post.tex	code/stata/15_descriptive_tables_figures.do	
Table A5	output/tables/main_effects_ts_span_math.tex	code/stata/16_main_effects_and_rob.do	
Table A6	output/tables/main_effects_ts_GPA_controls_yn.tex	code/stata/16_main_effects_and_rob.do	
Table A7	output/tables/descriptive_switches.tex	code/stata/20_identifying.do	
Table A8	output/tables/main_effects_ts_st_2025_v2.tex	code/stata/16_main_effects_and_rob.do	
Table A9	output/tables/het_effects_st_ts_GPA_bysimce.tex	code/stata/17_het_effects.do	
Table A10	output/tables/het_effects_ts_deciles_2025.tex	code/stata/17_het_effects.do	
Table A11	output/tables/het_effects_ts_allint.tex	code/stata/17_het_effects.do	
Table A12	output/tables/main_effects_ts_GPA_tsunami_2025.tex	code/stata/16_main_effects_and_rob.do	
Table A13	output/tables/main_effects_spatcor_ts_2025_v2.tex	code/stata/16_main_effects_and_rob.do	
Table A14	output/tables/conley_to_edit.tex + output/tables/Table_A14_conley.txt	code/stata/21_conley.do	Requires manual verification of Conley standard errors as described in Section 6.2
Table A15	output/tables/main_effects_ts_GPA_fe.tex	code/stata/16_main_effects_and_rob.do	
Table A16	output/tables/descriptive_missing_exp.tex	code/stata/19_school_resources.do	
Table A17	output/tables/med_effects_expenditures_select_attr.tex	code/stata/19_school_resources.do	
Table A18	output/tables/rendidosexp.tex	code/stata/19_school_resources.do	
Table A19	code/tex/Paper/appendix_reconstruction_policy.tex	—	Copy of Table from Gobierno de Chile, 2010, translated into English ; not generated from data.
Table A20	code/tex/Paper/appendix_reconstruction_policy.tex	—	Conceptual Table; not generated from data.

11.3 In-Text Numbers

Output in Paper	Output in Package	Script/Program to Execute	Comments
Section 2.1	output/in_text_numbers/Section_2_1.txt	code/stata/12_gen_variables.do	
Section 2.2, introduction	output/in_text_numbers/Section_2_2.txt	code/stata/14_analyze_census.do	
Sections 2.3 & 3, introduction (in-text numbers)	output/in_text_numbers/Sections_2_3.txt	code/stata/15_descriptive_tables_figures.do	
Section 4.1.1 (in-text numbers)	output/in_text_numbers/Section_4_1_1.txt	code/stata/16_main_effects_and_rob.do	
Section 4.1.2 (in-text numbers)	output/in_text_numbers/Section_4_1_2.txt	code/stata/18_med_relationships.do	
Section 4.1.2 (in-text confidence intervals)	output/tables/med_effects_curriculum_CIs_2025.tex	code/stata/18_med_relationships.do	
Section 4.1.3, introduction (in-text numbers)	output/in_text_numbers/Section_4_1_3.txt	code/stata/19_school_resources.do	

These text files allow direct verification of all numerical claims appearing in the manuscript.